



Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!) ○
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration Configure

Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` – installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements 1.

Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Student ID: 2024300753

Course/Year/Section: BSCS 3

Date Submitted: September 3, 2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS) Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu OS
Virtualization Software (VirtualBox/VMware)	Oracle Virtual Box
Ubuntu Server Version	Ubuntu-26.04-live-server-amd64
VM Name	TerranovaVMAkhylle
RAM Allocated	6 GB
Storage Allocated	100 GB
Network Mode	NAT
SSH Client Used	openssh

3 VM CONFIGURATION

Setting	Value
VM Name	TerranovaServerAkhylle
OS Type	Ubuntu (64-bit)
Memory	6000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION Commands

Executed:

- `sudo systemctl status ssh`
- `sudo apt update`
- `sudo apt install openssh-server -y`
- `sudo systemctl enable ssh`
- `sudo systemctl start ssh`
- `ip a`

- ssh Terranova@192.168.137.113

Server IP Address: 192.168.137.113

Network Interface: Inter(R) Dual Band Wireless-AC 7625 #2

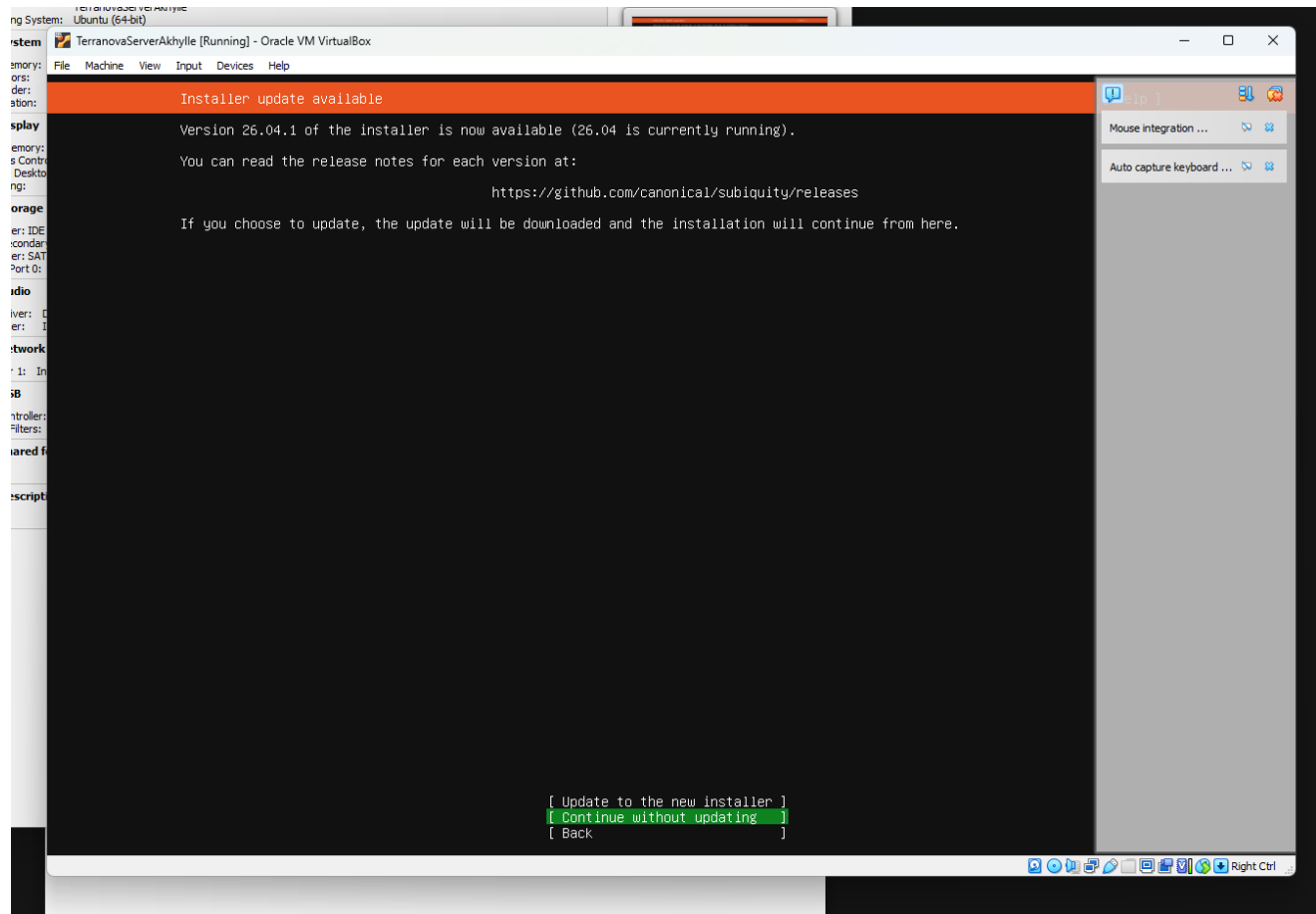
6. REMOTE CONNECTION Connection

Details:

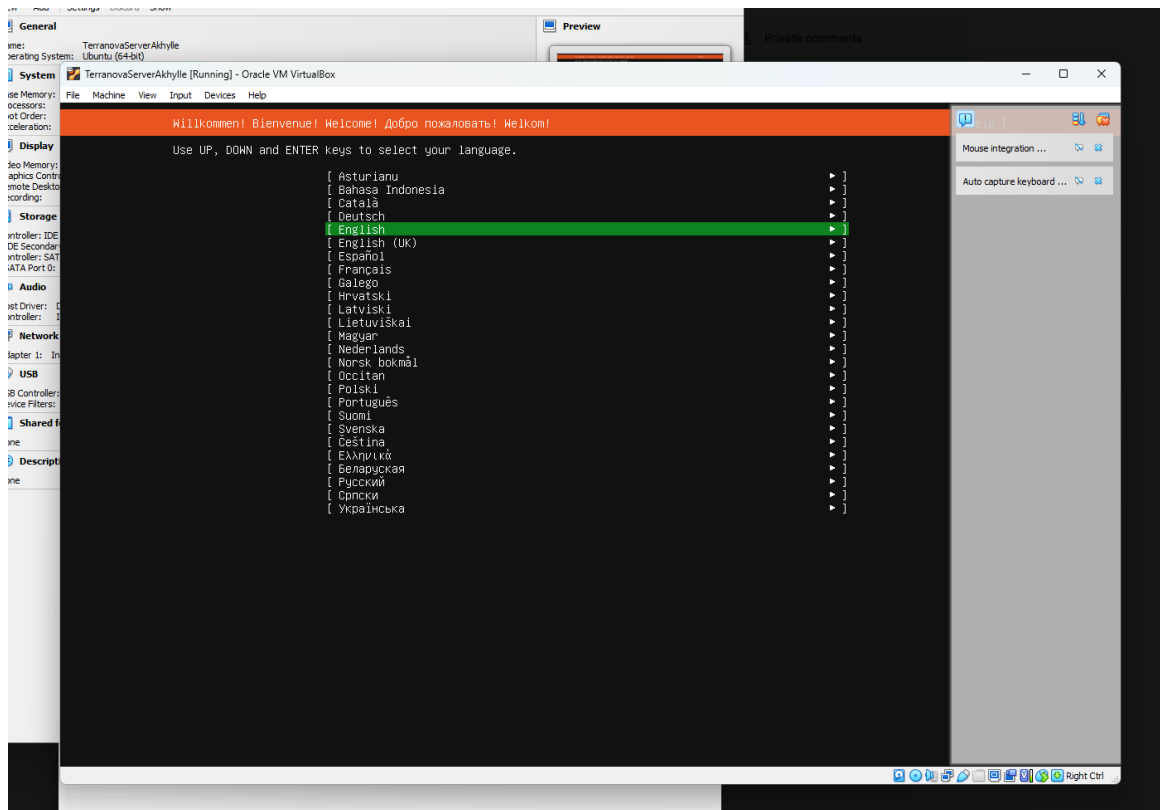
Parameter	Value
Server IP	192.168.137.113
Username	Terranova
SSH Port	22

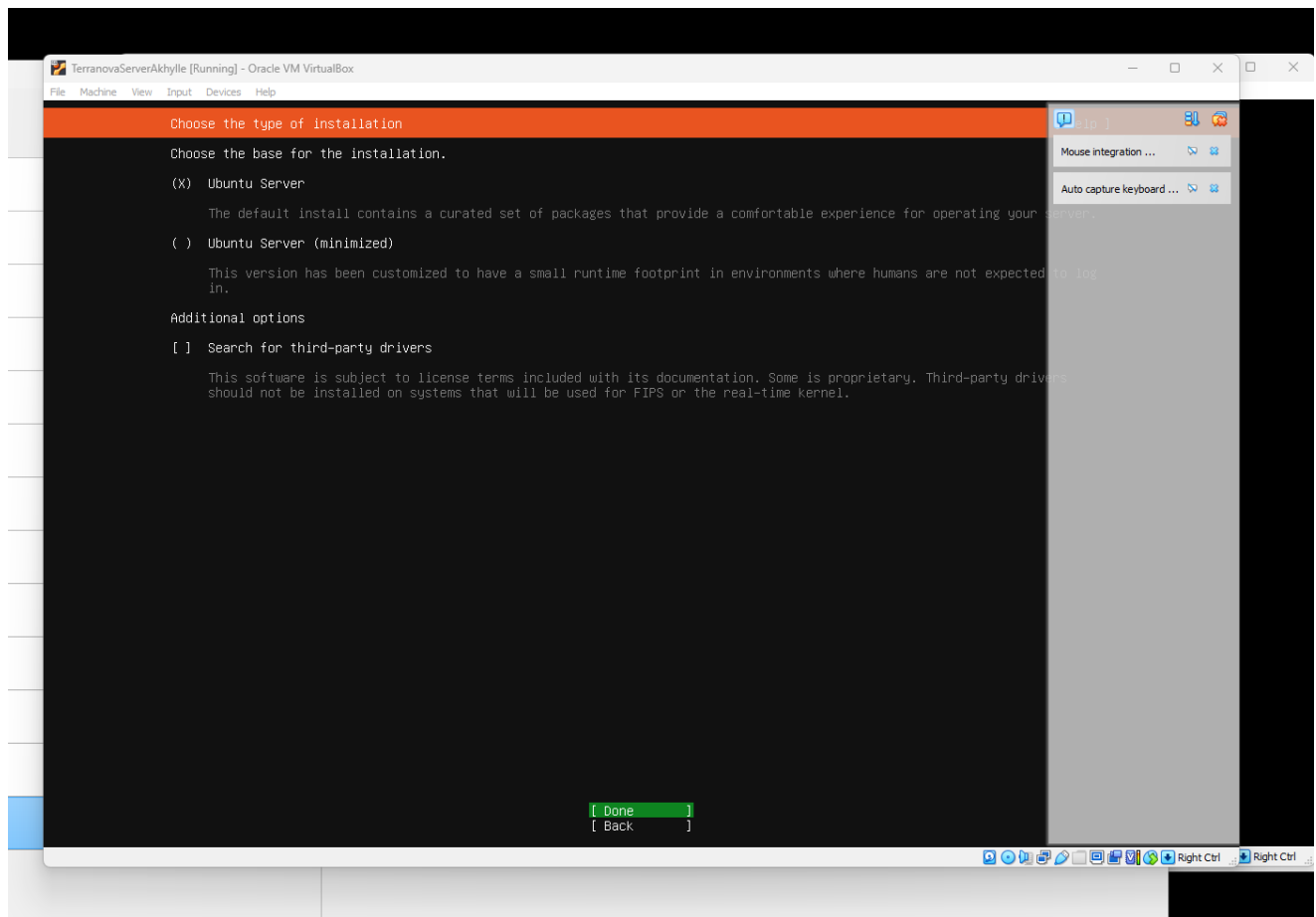
Screenshots

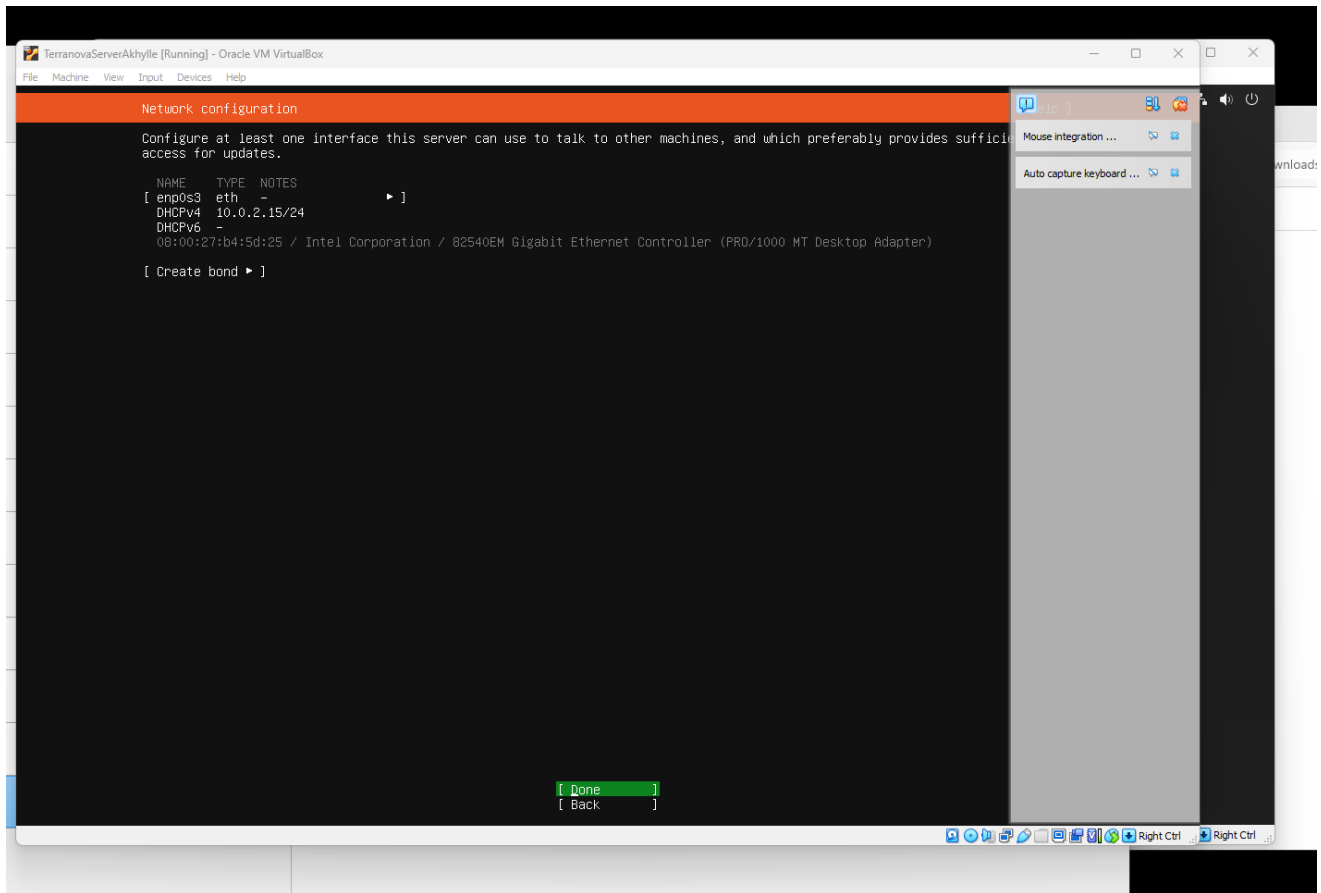
VM Server Installation



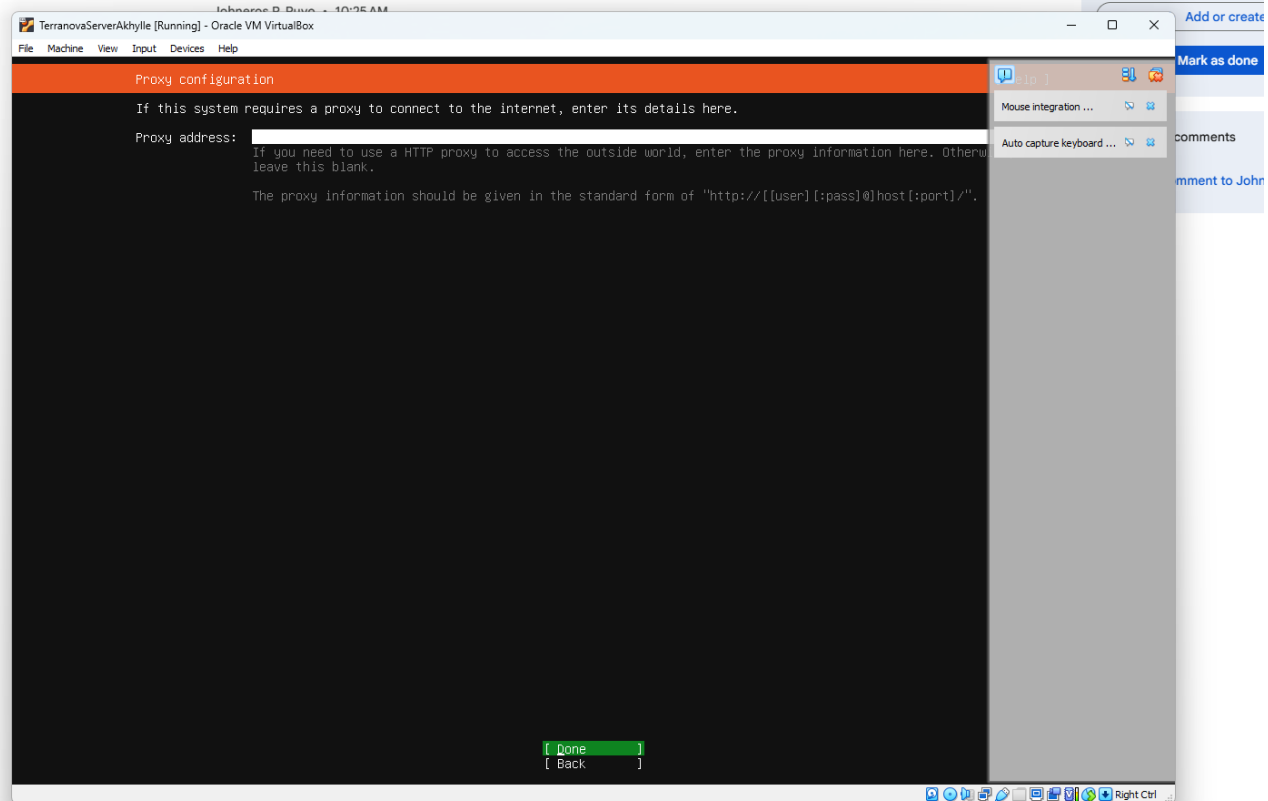
VM Server Prep







LAB 2 - VIRTUAL SERVER



LAB 2 - VIRTUAL SERVER

TerranovaServerAkhyle [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Ubuntu archive mirror configuration

If you use an alternative mirror for Ubuntu, enter its details here.

Mirror address:

You may provide an archive mirror to be used instead of the default.

This mirror location passed tests.

```
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Get:3 file:/cdrom/resolute Release.gpg [228 B]
Get:3 file:/cdrom/resolute Release.gpg [228 B]
Get:4 http://ph.archive.ubuntu.com/ubuntu/resolute InRelease [136 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu/resolute-updates InRelease [137 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu/resolute-backports InRelease [137 kB]
Fetched 409 kB in 2s (240 kB/s)
Reading package lists...
```

[Done]
[Back]

Mouse integration ...
Auto capture keyboard ...

Your work

Add or create

Mark as done

comments

comment to Johnner

LAB 2 - VIRTUAL SERVER

TerranovaServerAkhyle [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Guided storage configuration

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[VBOX_HARDDISK_VB0ae4bf14-55508b37 local disk 100.000G ▼]

(X) Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

[Done]
[Back]

Mouse integration ...
Auto capture keyboard ...

Your work

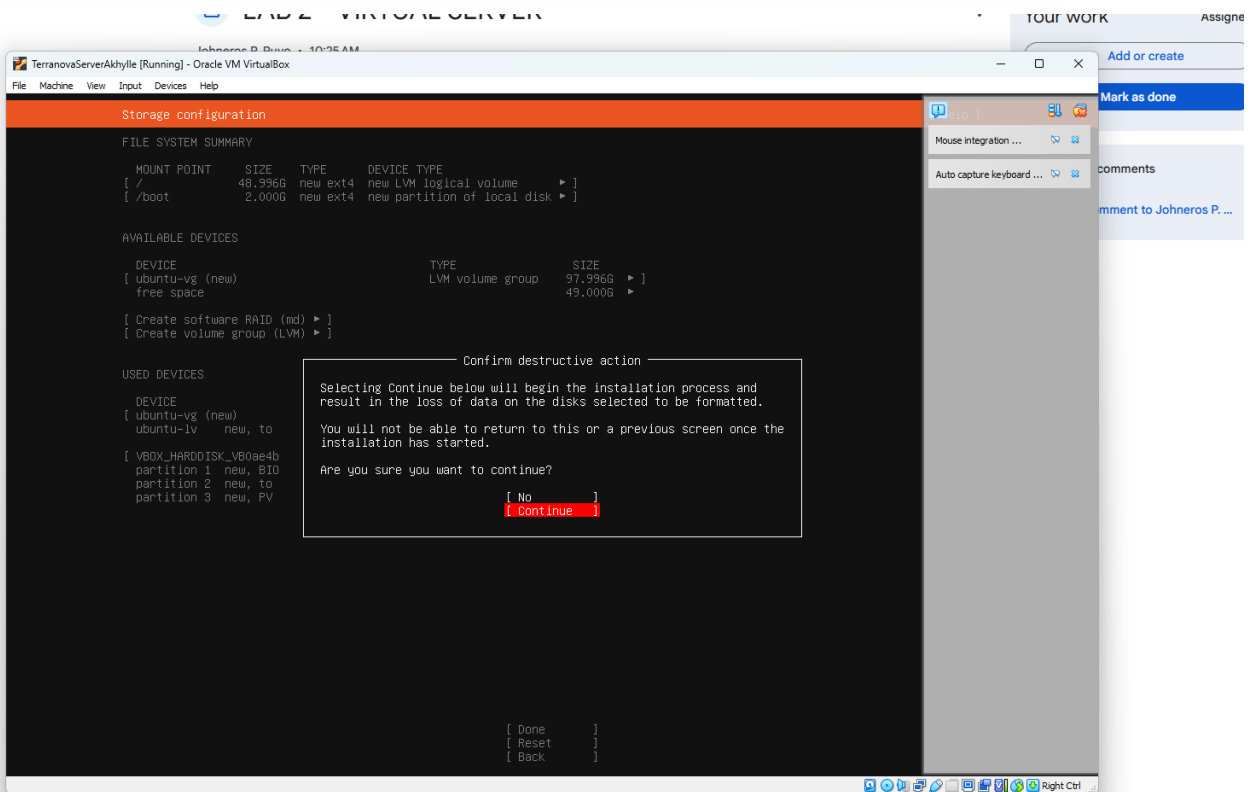
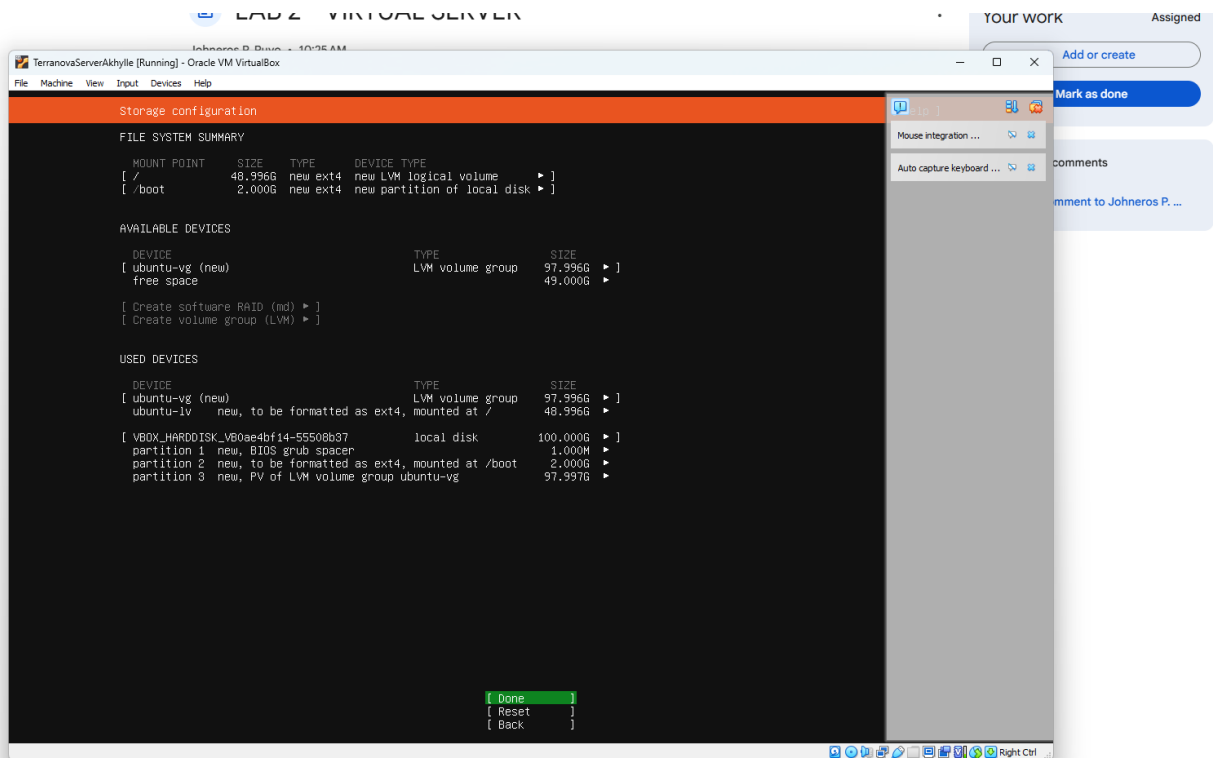
Assigned

Add or create

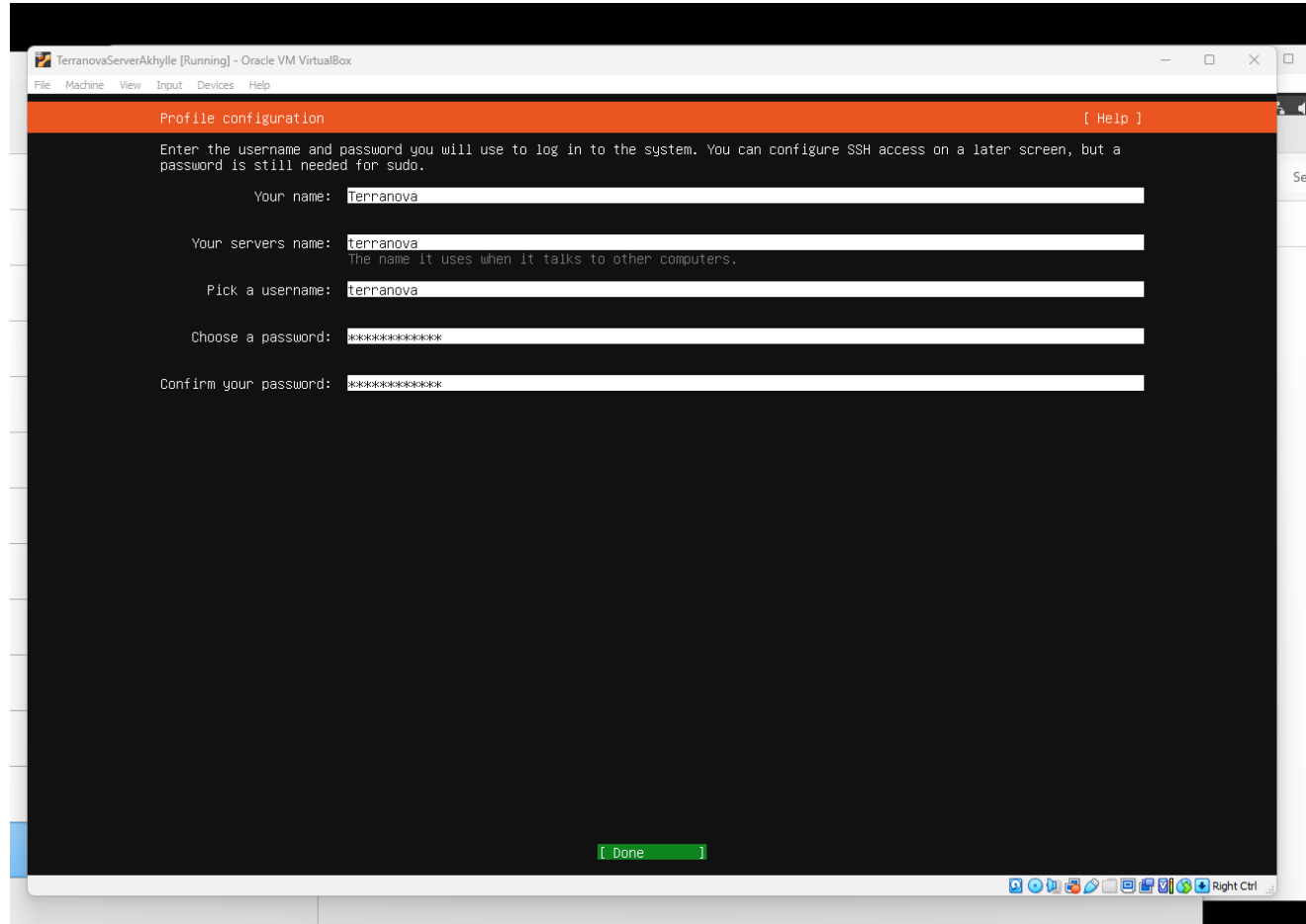
Mark as done

comments

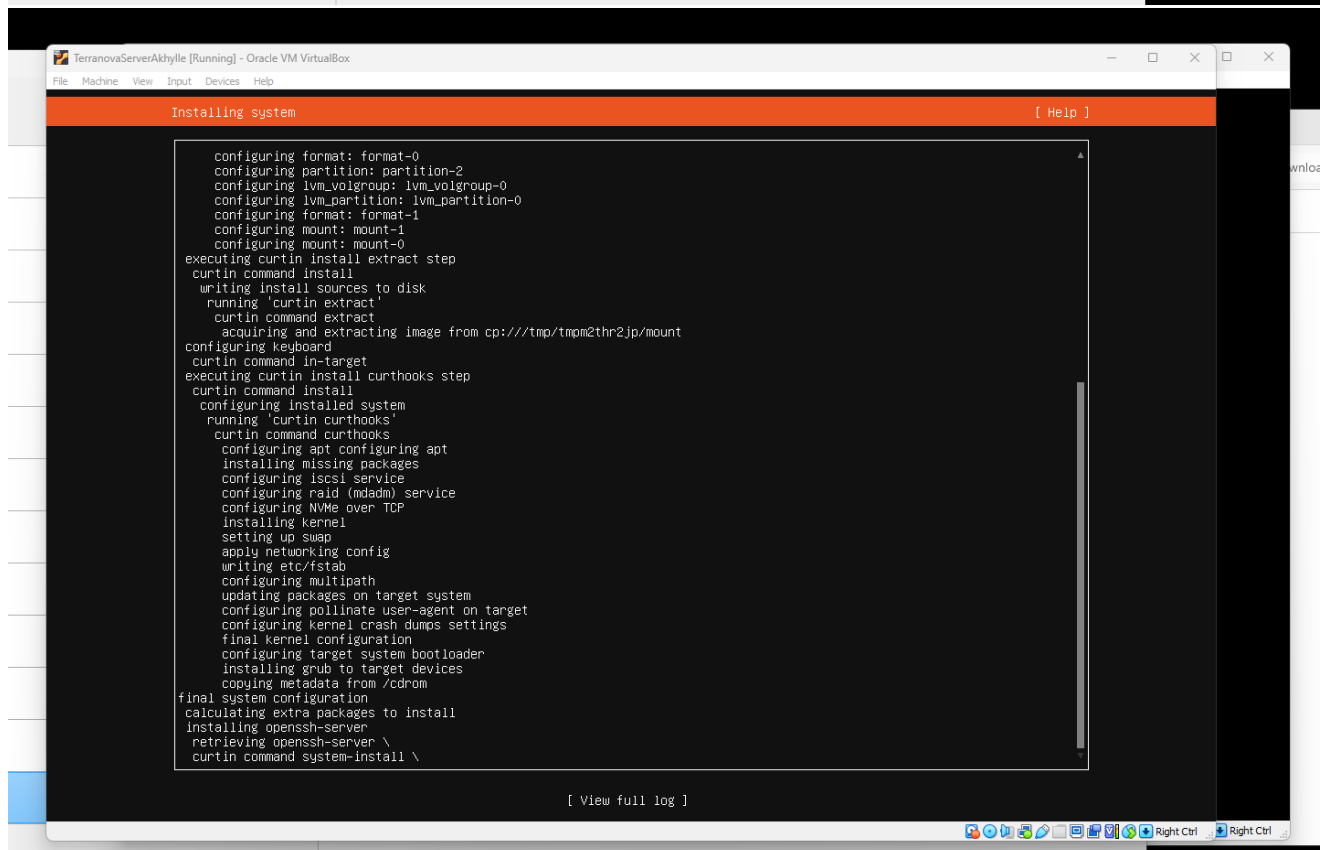
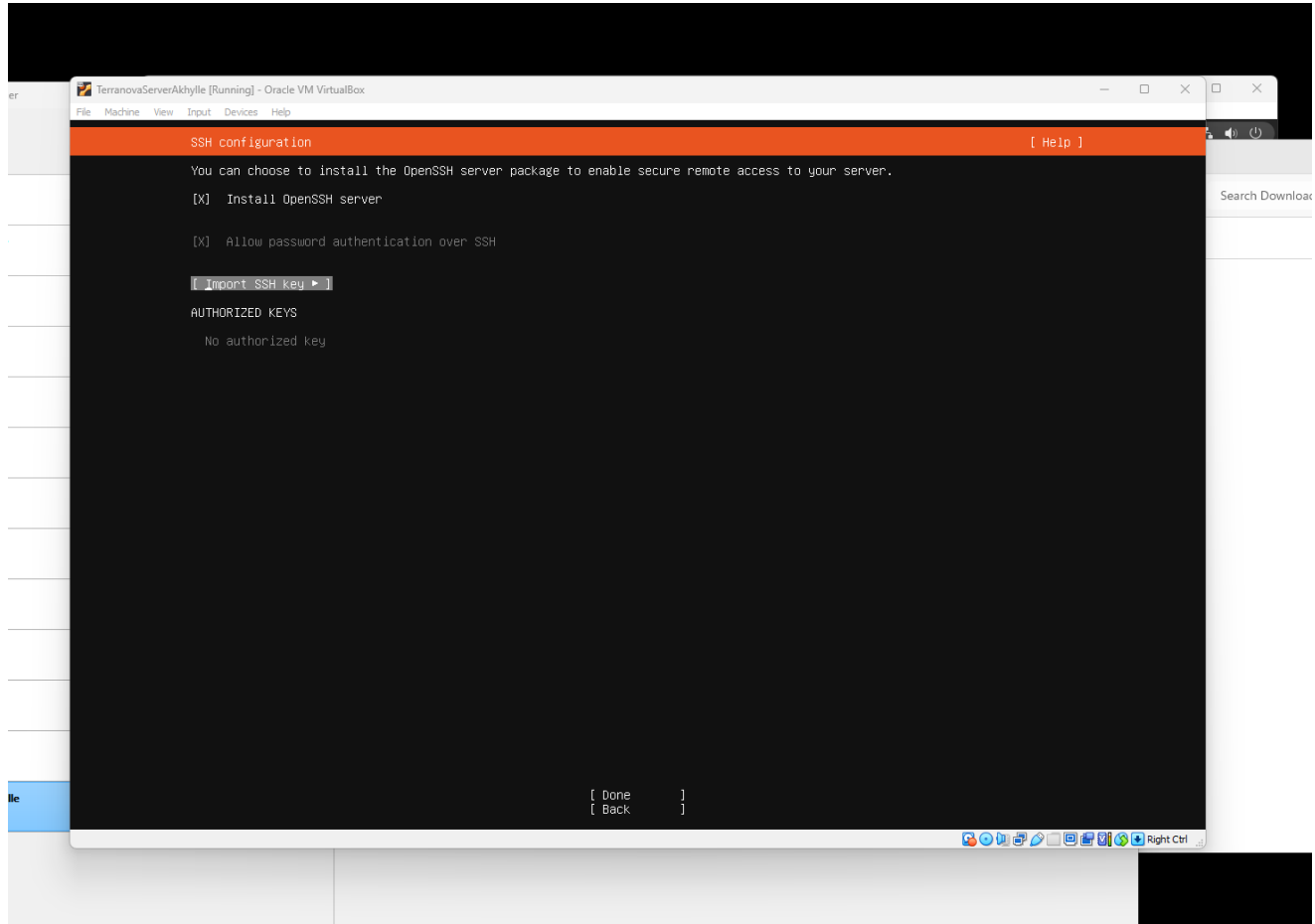
comment to Johnner P. ...



Profile Configuration



Install SSH Server



TerranovaServerAkhyllie [Running] - Oracle VM VirtualBox

Updating system [Help]

```
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpm2thr2jp/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
```

[View full log]

TerranovaServerAkhyllie [Running] - Oracle VM VirtualBox

Installation complete! [Help]

```
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpm2thr2jp/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtins:
subiquity/late/run_user_supplied:
```

[View full log]
[Reboot Now]

Server Prep

```
terranova login: terranova
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Sep  3 02:54:38 AM UTC 2026

System load:  0.12               Processes:    155
Usage of /:   14.9% of 47.9GB    Users logged in: 0
Memory usage: 4%                IPv4 address for enp0s3: 10.0.2.15
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

terranova@terranova:~$
```

```
TerranovaServer@kyle (Running) - Oracle VM VirtualBox
File  Machine  View  Input  Devices  Help

Login incorrect

terranova login: terranova
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

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 * Support:        https://ubuntu.com/pro

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Memory usage: 4%                IPv4 address for enp0s3: 10.0.2.15
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applicable law.

terranova@terranova:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:27:14:5d:25 brd ff:ff:ff:ff:ff:ff
    altname enp0s3v4-SGCS
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86107sec preferred_lft 86107sec
    inet6 fe80::a00:2714:5d25:64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
terranova@terranova:~$
```

```
terranova@terranova:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 02:53:14 UTC; 22min ago
   Invocation: c47838c2136a4655be68bc9c2bafacb5
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1369 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1372 (sshd)
     Tasks: 1 (limit: 3768)
    Memory: 1.6M (peak: 2.6M)
       CPU: 17ms
   CGroup: /system.slice/ssh.service
           └─1372 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 02:53:14 terranova systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 02:53:14 terranova sshd[1372]: Server listening on 0.0.0.0 port 22.
Sep 03 02:53:14 terranova sshd[1372]: Server listening on :: port 22.
Sep 03 02:53:14 terranova systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
terranova@terranova:~$
```

```

Sep 03 02:53:14 terranova systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
terranova@terranova:~$ sudo apt update
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
58 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
terranova@terranova:~$
```

```

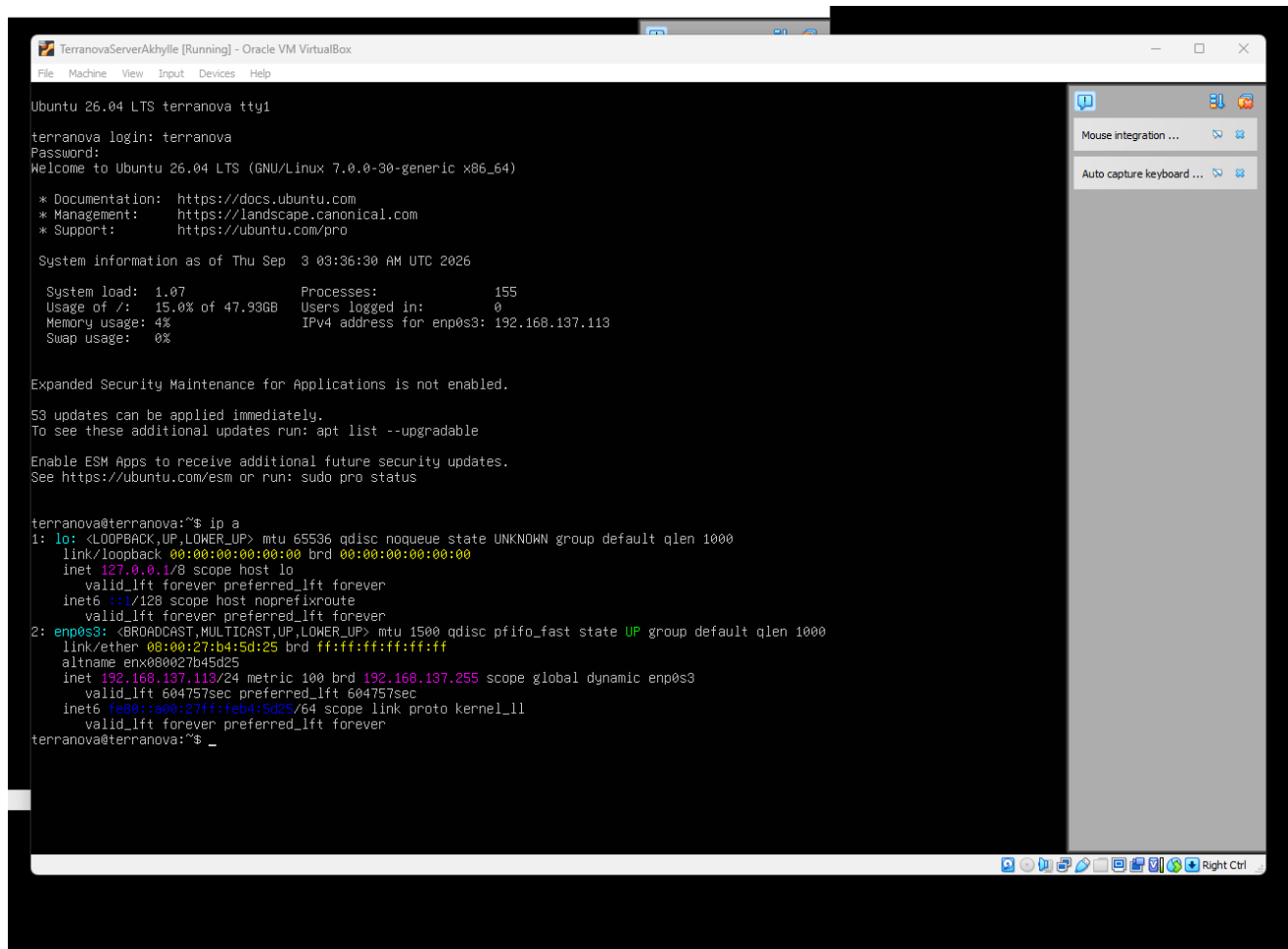
Error: Unable to locate package openssh-server-y
terranova@terranova:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 58
terranova@terranova:~$
```

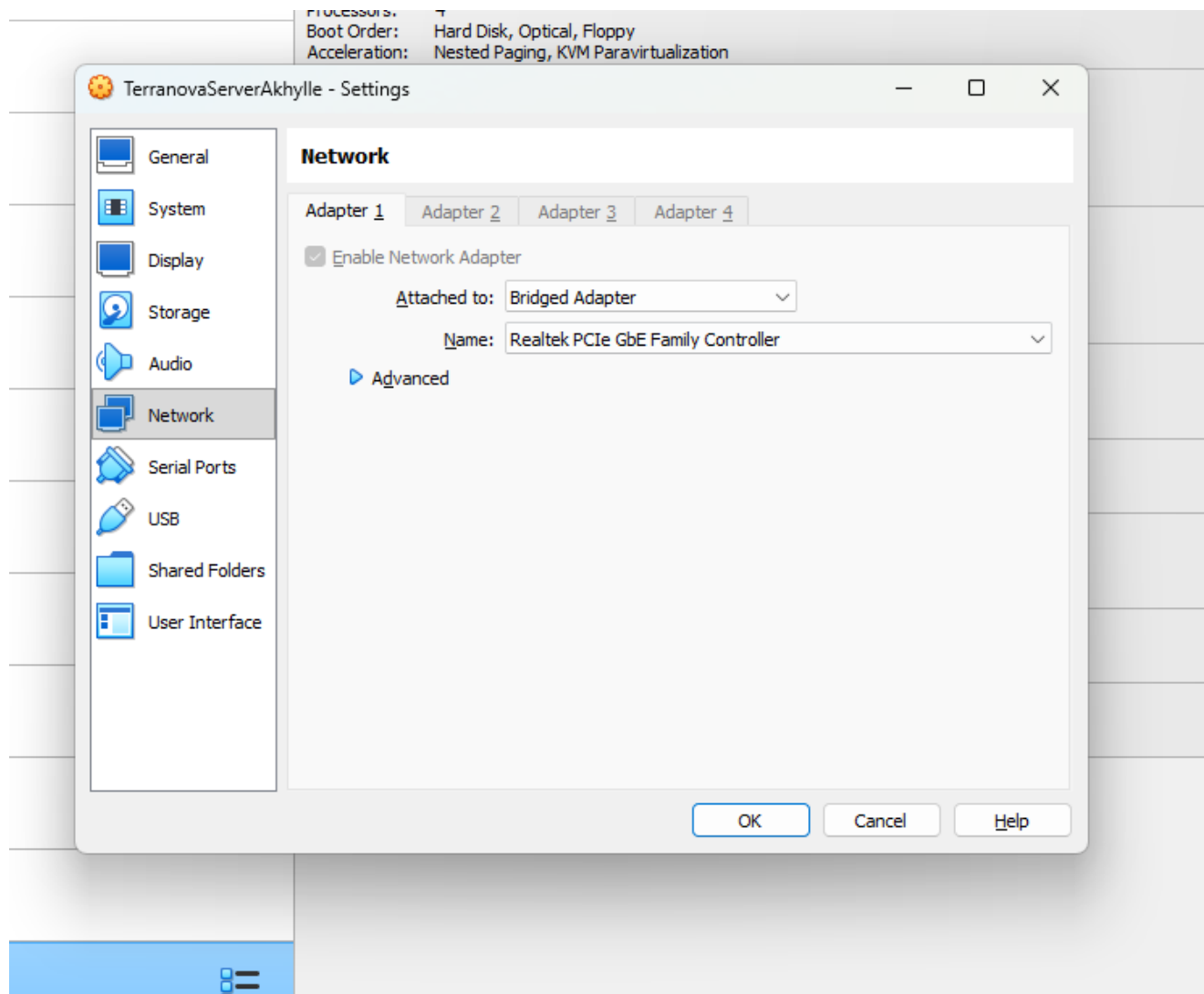
```

TerranovaServerAkhyll [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
58 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
terranova@terranova:~$ sudo apt install openssh-server-y
Error: Unable to locate package openssh-server-y
terranova@terranova:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 58
terranova@terranova:~$
a      install      openssh-server  ssh      sudo      update
apt     ip      openssh-server-y  status   systemctl  -y
terranova@terranova:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
terranova@terranova:~$ sudo systemctl start ssh
terranova@terranova:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b4:5d:25 brd ff:ff:ff:ff:ff:ff
    altname enx000027b45d25
    inet6 fe80::a00:27ff:feb4:5d25/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
terranova@terranova:~$
```



Network Configuration



Desktop Connection to the Server

```
terrano@192.168.137.113: Permission denied (publickey,password).
terrano@terrano-VirtualBox:~$ ssh terrano@192.168.137.113
terrano@192.168.137.113's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 03:39:41 AM UTC 2026

System load:  0.06               Processes:            159
Usage of /:   15.0% of 47.93GB   Users logged in:     0
Memory usage: 5%                IPv4 address for enp0s3: 192.168.137.113
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```